

EXPERIMENT 2

TITLE

8-Queen Problem using Backtracking Algorithm

AIM

To write a Python program to solve the **8-Queen Problem** using the **Backtracking Algorithm**.

OBJECTIVE

- To understand the concept of the 8-Queen problem.
- To implement the Backtracking algorithm.
- To place eight queens on a chessboard so that no two queens attack each other.
- To demonstrate constraint satisfaction in Artificial Intelligence.

ALGORITHM

Step 1: Create an empty 8×8 chessboard.

Step 2: Start placing queens from the first column.

Step 3: Check whether the current position is safe.

Step 4: If the position is safe, place the queen.

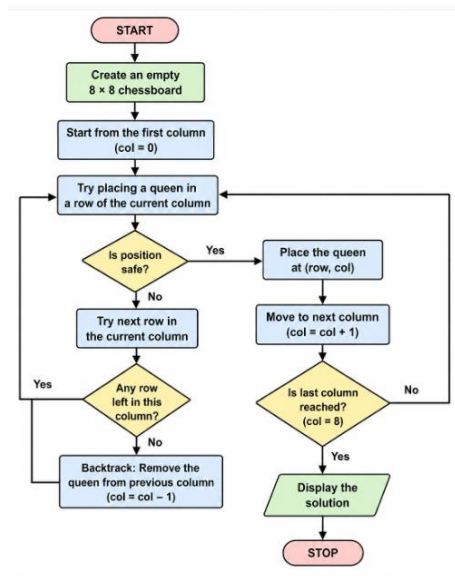
Step 5: Move to the next column and repeat the process.

Step 6: If no safe position is available, remove the previously placed queen (Backtracking).

Step 7: Continue until all eight queens are placed.

Step 8: Display the solution.

FLOWCHART



PYTHON CODE

N = 8

```
def print_board(board):
    for row in board:
        print(" ".join(row))

def is_safe(board, row, col):
    # Left Side
    for i in range(col):
        if board[row][i] == "Q":
            return False

    # Upper Diagonal
    i = row
    j = col
    while i >= 0 and j >= 0:
        if board[i][j] == "Q":
            return False
        i -= 1
        j -= 1

    # Lower Diagonal
    i = row
    j = col
    while i < N and j >= 0:
        if board[i][j] == "Q":
            return False
        i += 1
        j -= 1

    return True

def solve(board, col):
```

```

    if col >= N:
        return True

    for row in range(N):

        if is_safe(board, row, col):

            board[row][col] = "Q"

            if solve(board, col + 1):
                return True

            board[row][col] = "."

    return False

board = [ "." for _ in range(N) ] for _ in range(N)

if solve(board, 0):
    print("Solution Found:\n")
    print_board(board)
else:
    print("No Solution")

```

SAMPLE OUTPUT

Solution Found:

```

Q . . . . . . .
. . . . Q . . .
. . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .

```

CONCLUSION

The **8-Queen Problem** was solved successfully using the **Backtracking Algorithm**. The algorithm places queens safely by checking conflicts and uses backtracking whenever a conflict occurs, ensuring that all eight queens are placed without attacking each other.